MAIN TITLE

Abstract.

1. Introduction

2. Active oscillator–actuator model

Active vibration control combines structural sensing, feedback control, and force actuation to modify the dynamic response of lightly damped mechanical structures. Direct velocity feedback is widely used because of its simplicity and effectiveness in reducing resonant vibrations [1, 7, 4]. In practical implementations, however, actuators possess finite bandwidth and internal dynamics, which may substantially modify the stability of the closed-loop system [3]. Moreover, actuator nonlinearities may produce self-excited oscillations and limit cycles [2].

Motivated by these observations, we consider the reduced-order oscillator–actuator model

$$\begin{aligned} \dot{x} &= v, \\ \dot{v} &= -\omega_0^2 x - (d-g)v + f, \\ \tau \dot{f} &= k_s x - f - \kappa x^2. \end{aligned} \tag{1}$$

Here x denotes the modal displacement, $v = \dot{x}$ is the modal velocity, and f is the actuator force per unit modal mass. The parameter $\omega_0 > 0$ is the natural frequency of the uncontrolled structure, $d > 0$ is the passive damping coefficient, g is the velocity-feedback gain, $\tau > 0$ is the actuator time constant, and k_s is the linear stiffness-compensation gain. The coefficient $\kappa \neq 0$ represents the leading nonlinear actuator distortion.

The actuator equation can be written as the first-order tracking law

$$\tau \dot{f} + f = f_c(x), \qquad f_c(x) = k_s x - \kappa x^2.$$

When $k_s = \omega_0^2$, the linear actuator force exactly compensates the structural stiffness. The parameter k_s also provides a natural second unfolding direction for the zero–Hopf singularity.

2.1. Equilibria and the zero–Hopf point.

The equilibrium equations associated with (1) are

$$v = 0, \qquad f = \omega_0^2 x,$$

and

$$k_s x - f - \kappa x^2 = 0.$$

Substitution of $f = \omega_0^2 x$ yields

$$(k_s - \omega_0^2)x - \kappa x^2 = 0. \tag{2}$$

Consequently, the system possesses the equilibrium branches

$$E_0 = (0, 0, 0) \tag{3}$$

and

$$E_1 = \left(\frac{k_s - \omega_0^2}{\kappa}, 0, \frac{\omega_0^2 (k_s - \omega_0^2)}{\kappa} \right). \tag{4}$$

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At the critical stiffness-compensation value $k_s = \omega_0^2$, the two equilibrium branches coalesce at the origin.

The Jacobian matrix of system (1) at E_0 is

$$A(g, k_s) = \begin{pmatrix} 0 & 1 & 0 \\ -\omega_0^2 & g - d & 1 \\ k_s/\tau & 0 & -1/\tau \end{pmatrix}. \tag{5}$$

Its characteristic polynomial is

$$p(\lambda; g, k_s) = \lambda^3 + \left(d - g + \frac{1}{\tau}\right)\lambda^2 + \left(\omega_0^2 + \frac{d-g}{\tau}\right)\lambda + \frac{\omega_0^2 - k_s}{\tau}. \tag{6}$$

The origin is a zero–Hopf equilibrium when the constant term and the quadratic coefficient in (6) vanish, while the coefficient of λ remains positive. Hence the zero–Hopf critical parameters are

$$g_{\mathrm{ZH}} = d + \frac{1}{\tau}, \qquad k_{s,\mathrm{ZH}} = \omega_0^2. \tag{7}$$

At these values,

$$p_{\mathrm{ZH}}(\lambda) = \lambda\left[\lambda^2 + \left(\omega_0^2 - \frac{1}{\tau^2}\right)\right].$$

Assume

$$\omega_0 \tau > 1 \tag{8}$$

and define the zero–Hopf oscillatory frequency by

$$\Omega_{\mathrm{ZH}} = \sqrt{\omega_0^2 - \frac{1}{\tau^2}}. \tag{9}$$

Then $\Omega_{\mathrm{ZH}} > 0$, and the eigenvalues at the zero–Hopf point are

$$\lambda_1 = 0, \qquad \lambda_{2,3} = \pm i\Omega_{\mathrm{ZH}}. \tag{10}$$

We study the one-parameter unfolding

$$g = g_{\mathrm{ZH}} + g_1\varepsilon = d + \frac{1}{\tau} + g_1\varepsilon, \qquad k_s = k_{s,\mathrm{ZH}} + k_1\varepsilon = \omega_0^2 + k_1\varepsilon, \tag{11}$$

where $g_1, k_1 \in \mathbb{R}$ are fixed and $0 < |\varepsilon| \ll 1$. This path represents a one-parameter slice through the codimension-two zero–Hopf point in the (g, k_s)-parameter plane.

2.2. **Main Results.**

Theorem 2.1 (Zero-Hopf bifurcation)**.** *Consider system* (1)*. Assume*

$$\omega_0 > 0, \qquad \tau > 0, \qquad \kappa \neq 0, \qquad \omega_0\tau > 1,$$

and let $g = d + \frac{1}{\tau} + g_1\varepsilon$ *and* $k_s = \omega_0^2 + k_1\varepsilon$*. If*

$$k_1^2 - g_1^2\Omega_{\mathrm{ZH}}^4\tau^2 > 0, \tag{12}$$

then, for every sufficiently small $\varepsilon \neq 0$*, system* (1) *possesses a periodic orbit bifurcating from the zero–Hopf equilibrium* E_0*. Its period satisfies*

$$T_\varepsilon = \frac{2\pi}{\Omega_{\mathrm{ZH}}} + O(\varepsilon). \tag{13}$$

Moreover, the periodic orbit is orbitally asymptotically stable if $\varepsilon g_1 < 0$ *and unstable if* $\varepsilon g_1 > 0$*. The case* $g_1 = 0$ *cannot be decided by first-order averaging.*

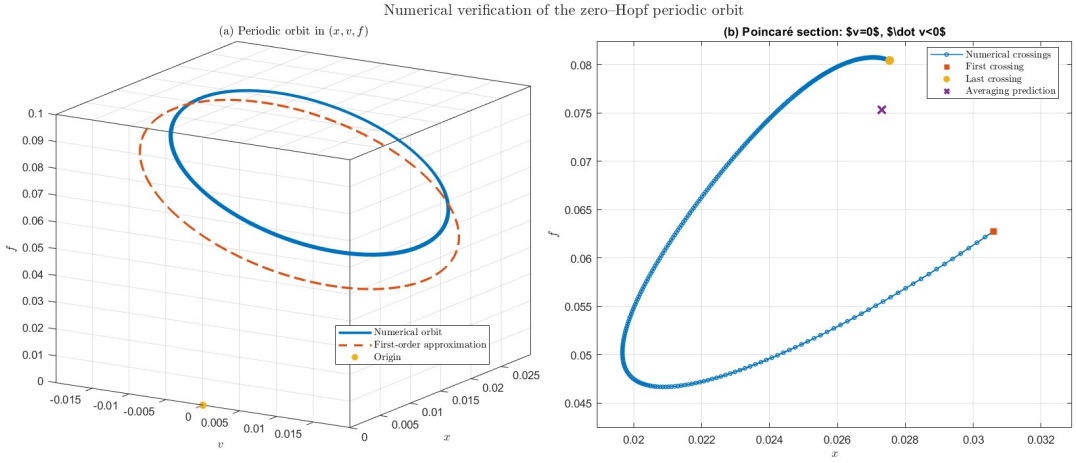


FIGURE 1. Numerical verification of the periodic orbit bifurcating from the zero–Hopf equilibrium for $\omega_0 = 2$, $\tau = 1$, $\kappa = 1$, $d = 0.2$, $g_1 = -0.2$, $k_1 = 1$, and $\varepsilon = 0.02$. Panel (a) compares the periodic orbit obtained by numerical integration of the full oscillator–actuator system with its leading-order approximation obtained by averaging. Panel (b) shows successive intersections of the numerical trajectory with the Poincaré section $v = 0$, $\dot v < 0$. Their convergence toward a single point provides numerical evidence for the orbital asymptotic stability predicted by $\varepsilon g_1 < 0$.

Theorem 2.2 (Hopf bifurcation)**.** *Consider system* (1) *with*

$$\omega_0 > 0, \qquad \tau > 0, \qquad \kappa \neq 0,$$

and fix $\sigma \neq 0$ *such that* $\Omega^2 := \omega_0^2 - \tau^{-2} + \frac{\sigma}{\tau} > 0$. *Set*

$$g_{\mathrm{H}} = d + \tau^{-1} - \sigma, \qquad k_{s,\mathrm{H}} = \omega_0^2 - \tau\sigma\Omega^2.$$

Then, for $g = g_{\mathrm{H}}$, *the origin is a nondegenerate Hopf equilibrium at* $k_s = k_{s,\mathrm{H}}$, *with eigenvalues* $-\sigma, \pm i\Omega$. *Assume*

$$k_s = k_{s,\mathrm{H}} + \eta_2\varepsilon^2 + \eta_3\varepsilon^3, \qquad \sigma\eta_2 < 0,$$

where $\eta_2, \eta_3 \in \mathbb{R}$ *and* $0 < |\varepsilon| \ll 1$. *Then a small-amplitude periodic orbit bifurcates from the origin. Its displacement amplitude and period satisfy*

$$A_\varepsilon = |\varepsilon| R^*_{\mathrm{H}} + O(\varepsilon^2), \qquad T_\varepsilon = \frac{2\pi}{\Omega} + O(\varepsilon^2),$$

where

$$R^*_{\mathrm{H}} = \left[-\frac{2\tau\Omega^2\sigma(4\Omega^2 + \sigma^2)}{\kappa^2(8\Omega^2 + \sigma^2)}\,\eta_2 \right]^{1/2}.$$

Moreover, the bifurcation is supercritical and the periodic orbit is orbitally asymptotically stable when $\sigma > 0$, *whereas it is subcritical and the periodic orbit is unstable when* $\sigma < 0$.

Theorem 2.3 (Dynamics on the sphere at infinity)**.** *Consider system* (1) *with* $\omega_0 > 0$, $\tau > 0$, *and* $\kappa \neq 0$. *Its degree-two Poincaré compactification has the following boundary dynamics.*

(1) The equilibrium set is the great circle $\mathcal{E}_\infty = \{X = 0\} \cap S^2_\infty$, *and the boundary linearization vanishes at every equilibrium. Away from* $p_f^\pm$, *the local portrait consists of an equilibrium line with parallel moving trajectories on both sides. At* $p_f^\pm$, *it consists of an equilibrium line and incoming or outgoing rays in the force charts.*

*(2) Every nonstationary orbit is a great semicircle (**??**), directed from p_f^+ to p_f^- if $\kappa > 0$, and oppositely if $\kappa < 0$. There are no nonconstant periodic or homoclinic boundary orbits.*

(3) The common forward-limit endpoint is Lyapunov stable but not asymptotically stable on the sphere. The other endpoint is unstable forward in time and Lyapunov stable in reversed time. The oriented boundary portrait depends only on $\operatorname{sgn}\kappa$.

2.3. **Real canonical coordinates at the zero–Hopf point.** At the critical parameter values

$$g = g_{\mathrm{ZH}} = d + \frac{1}{\tau}, \qquad k_s = k_{s,\mathrm{ZH}} = \omega_0^2,$$

the Jacobian matrix at the origin is

$$A_{\mathrm{ZH}} = \begin{pmatrix} 0 & 1 & 0 \\ -\omega_0^2 & 1/\tau & 1 \\ \omega_0^2/\tau & 0 & -1/\tau \end{pmatrix}. \tag{14}$$

We seek a real nonsingular matrix P_{ZH} satisfying

$$P_{\mathrm{ZH}}^{-1} A_{\mathrm{ZH}} P_{\mathrm{ZH}} = J_{\mathrm{ZH}}, \tag{15}$$

where

$$J_{\mathrm{ZH}} = \begin{pmatrix} 0 & -\Omega_{\mathrm{ZH}} & 0 \\ \Omega_{\mathrm{ZH}} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{16}$$

Writing

$$P_{\mathrm{ZH}} = \begin{pmatrix} p_1 & p_2 & p_3 \end{pmatrix},$$

its columns are chosen to satisfy

$$A_{\mathrm{ZH}} p_1 = \Omega_{\mathrm{ZH}} p_2, \qquad A_{\mathrm{ZH}} p_2 = -\Omega_{\mathrm{ZH}} p_1, \qquad A_{\mathrm{ZH}} p_3 = 0. \tag{17}$$

A convenient choice is

$$P_{\mathrm{ZH}} = \begin{pmatrix} 1 & 0 & 1/\omega_0^2 \\ 0 & -\Omega_{\mathrm{ZH}} & 0 \\ 1/\tau^2 & \Omega_{\mathrm{ZH}}/\tau & 1 \end{pmatrix}. \tag{18}$$

Its determinant is

$$\det P_{\mathrm{ZH}} = -\frac{\Omega_{\mathrm{ZH}}^3}{\omega_0^2} \neq 0. \tag{19}$$

Thus P_{ZH} is nonsingular.

We introduce the real zero–Hopf coordinates (X, Y, Z) by

$$\begin{pmatrix} x \\ v \\ f \end{pmatrix} = P_{\mathrm{ZH}} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}. \tag{20}$$

Equivalently,

$$\begin{aligned} x &= X + \frac{Z}{\omega_0^2}, \\ v &= -\Omega_{\mathrm{ZH}} Y, \\ f &= \frac{X}{\tau^2} + \frac{\Omega_{\mathrm{ZH}}}{\tau} Y + Z. \end{aligned} \tag{21}$$

Under the unfolding (11), system (1) becomes

$$\begin{aligned}
\dot{X} =& -\Omega_{\mathrm{ZH}} Y + \varepsilon \left[\frac{g_1}{\Omega_{\mathrm{ZH}}\tau} Y - \frac{k_1}{\Omega_{\mathrm{ZH}}^2 \tau} X - \frac{k_1}{\Omega_{\mathrm{ZH}}^2 \omega_0^2 \tau} Z \right] \\
&+ \frac{\kappa}{\Omega_{\mathrm{ZH}}^2 \tau} X^2 + \frac{2\kappa}{\Omega_{\mathrm{ZH}}^2 \omega_0^2 \tau} XZ + \frac{\kappa}{\Omega_{\mathrm{ZH}}^2 \omega_0^4 \tau} Z^2, \\
\dot{Y} =& \Omega_{\mathrm{ZH}} X + \varepsilon g_1 Y, \\
\dot{Z} =& \varepsilon \left[-\frac{g_1 \omega_0^2}{\Omega_{\mathrm{ZH}}\tau} Y + \frac{k_1 \omega_0^2}{\Omega_{\mathrm{ZH}}^2 \tau} X + \frac{k_1}{\Omega_{\mathrm{ZH}}^2 \tau} Z \right] \\
&- \frac{\kappa \omega_0^2}{\Omega_{\mathrm{ZH}}^2 \tau} X^2 - \frac{2\kappa}{\Omega_{\mathrm{ZH}}^2 \tau} XZ - \frac{\kappa}{\Omega_{\mathrm{ZH}}^2 \omega_0^2 \tau} Z^2.
\end{aligned} \tag{22}$$

At $\varepsilon = 0$, the linear motion in the (X, Y)-plane is a harmonic rotation with angular frequency Ω_{ZH}, while Z is a neutral coordinate.

2.4. **Scaled polar formulation.** We seek periodic solutions whose distance from E_0 is $O(|\varepsilon|)$. Accordingly, introduce

$$X = \varepsilon r \cos\theta, \qquad Y = \varepsilon r \sin\theta, \qquad Z = \varepsilon u, \tag{23}$$

where

$$r \geq 0, \qquad u \in \mathbb{R}, \qquad \theta \in \mathbb{S}^1.$$

Define

$$\mathcal{H}(r, u, \theta) = r\cos\theta + \frac{u}{\omega_0^2}. \tag{24}$$

Then the original displacement variable satisfies

$$x = \varepsilon \mathcal{H}(r, u, \theta).$$

After substituting (23) into (22) and taking θ as the new independent variable, the system assumes the standard first-order averaging form

$$\begin{aligned}
r' &= \varepsilon F_{\mathrm{ZH},1}(\theta, r, u) + O(\varepsilon^2), \\
u' &= \varepsilon F_{\mathrm{ZH},2}(\theta, r, u) + O(\varepsilon^2),
\end{aligned} \tag{25}$$

where the prime denotes differentiation with respect to θ.

To simplify the notation, define

$$\begin{aligned}
\mathcal{P}(r, u, \theta) &= \frac{g_1 r \sin\theta}{\Omega_{\mathrm{ZH}}\tau} - \frac{k_1}{\Omega_{\mathrm{ZH}}^2 \tau} \mathcal{H}(r, u, \theta) + \frac{\kappa}{\Omega_{\mathrm{ZH}}^2 \tau} \mathcal{H}^2(r, u, \theta), \\
\mathcal{Q}(r, u, \theta) &= g_1 r \sin\theta, \\
\mathcal{R}(r, u, \theta) =& -\frac{g_1 \omega_0^2 r \sin\theta}{\Omega_{\mathrm{ZH}}\tau} + \frac{k_1 \omega_0^2}{\Omega_{\mathrm{ZH}}^2 \tau} \mathcal{H}(r, u, \theta) \\
&- \frac{\kappa \omega_0^2}{\Omega_{\mathrm{ZH}}^2 \tau} \mathcal{H}^2(r, u, \theta).
\end{aligned} \tag{26}$$

Then

$$F_{\mathrm{ZH},1} = \frac{\mathcal{P}\cos\theta + \mathcal{Q}\sin\theta}{\Omega_{\mathrm{ZH}}}, \qquad F_{\mathrm{ZH},2} = \frac{\mathcal{R}}{\Omega_{\mathrm{ZH}}}. \tag{27}$$

The functions $F_{\mathrm{ZH},1}$ and $F_{\mathrm{ZH},2}$ are 2π-periodic with respect to θ.

2.5. **First-order averaging.** The first averaged function associated with (25) is

$$\mathbf{f}_{\mathrm{ZH}}(r,u)=\begin{pmatrix} f_{\mathrm{ZH},1}(r,u)\\ f_{\mathrm{ZH},2}(r,u)\end{pmatrix}=\int_0^{2\pi}\begin{pmatrix} F_{\mathrm{ZH},1}(\theta,r,u)\\ F_{\mathrm{ZH},2}(\theta,r,u)\end{pmatrix}\,d\theta. \tag{28}$$

Using (26), the integrands can be written explicitly as

$$\begin{aligned} F_{\mathrm{ZH},1}=&\frac{g_1 r\sin\theta\cos\theta}{\Omega_{\mathrm{ZH}}^2\tau}+\frac{g_1 r\sin^2\theta}{\Omega_{\mathrm{ZH}}}-\frac{k_1 r\cos^2\theta}{\Omega_{\mathrm{ZH}}^3\tau}\\ &-\frac{k_1 u\cos\theta}{\Omega_{\mathrm{ZH}}^3\tau\omega_0^2}+\frac{\kappa r^2\cos^3\theta}{\Omega_{\mathrm{ZH}}^3\tau}\\ &+\frac{2\kappa r u\cos^2\theta}{\Omega_{\mathrm{ZH}}^3\tau\omega_0^2}+\frac{\kappa u^2\cos\theta}{\Omega_{\mathrm{ZH}}^3\tau\omega_0^4}, \end{aligned} \tag{29}$$

and

$$\begin{aligned} F_{\mathrm{ZH},2}=&-\frac{g_1\omega_0^2 r\sin\theta}{\Omega_{\mathrm{ZH}}^2\tau}+\frac{k_1\omega_0^2 r\cos\theta}{\Omega_{\mathrm{ZH}}^3\tau}+\frac{k_1 u}{\Omega_{\mathrm{ZH}}^3\tau}\\ &-\frac{\kappa\omega_0^2 r^2\cos^2\theta}{\Omega_{\mathrm{ZH}}^3\tau}-\frac{2\kappa r u\cos\theta}{\Omega_{\mathrm{ZH}}^3\tau}-\frac{\kappa u^2}{\Omega_{\mathrm{ZH}}^3\tau\omega_0^2}. \end{aligned} \tag{30}$$

Integrating over one complete period yields

$$f_{\mathrm{ZH},1}(r,u)=\frac{\pi r}{\Omega_{\mathrm{ZH}}}\left(g_1-\frac{k_1}{\Omega_{\mathrm{ZH}}^2\tau}+\frac{2\kappa u}{\Omega_{\mathrm{ZH}}^2\tau\omega_0^2}\right), \tag{31}$$

and

$$f_{\mathrm{ZH},2}(r,u)=\frac{\pi}{\Omega_{\mathrm{ZH}}^3\tau}\left(2k_1 u-\kappa\omega_0^2 r^2-\frac{2\kappa u^2}{\omega_0^2}\right). \tag{32}$$

Therefore,

$$\mathbf{f}_{\mathrm{ZH}}(r,u)=\begin{pmatrix} \dfrac{\pi r}{\Omega_{\mathrm{ZH}}}\left(g_1-\dfrac{k_1}{\Omega_{\mathrm{ZH}}^2\tau}+\dfrac{2\kappa u}{\Omega_{\mathrm{ZH}}^2\tau\omega_0^2}\right)\\[2ex] \dfrac{\pi}{\Omega_{\mathrm{ZH}}^3\tau}\left(2k_1 u-\kappa\omega_0^2 r^2-\dfrac{2\kappa u^2}{\omega_0^2}\right)\end{pmatrix}. \tag{33}$$

In particular, $\mathbf{f}_{\mathrm{ZH}}$ does not vanish identically. Thus first-order averaging is sufficient to detect periodic solutions bifurcating from the zero–Hopf equilibrium.

2.6. **Zeros of the averaged function.** We seek zeros

$$\mathbf{f}_{\mathrm{ZH}}(r,u)=0,\qquad r>0.$$

The condition $r>0$ excludes the equilibrium and corresponds to a genuine oscillatory solution.

Since $r>0$, the equation $f_{\mathrm{ZH},1}(r,u)=0$ is equivalent to

$$g_1-\frac{k_1}{\Omega_{\mathrm{ZH}}^2\tau}+\frac{2\kappa u}{\Omega_{\mathrm{ZH}}^2\tau\omega_0^2}=0.$$

Hence

$$u_0=\frac{\omega_0^2}{2\kappa}\left(k_1-g_1\Omega_{\mathrm{ZH}}^2\tau\right). \tag{34}$$

Substituting (34) into $f_{\mathrm{ZH},2}(r,u)=0$ gives

$$2k_1 u_0-\kappa\omega_0^2 r^2-\frac{2\kappa u_0^2}{\omega_0^2}=0.$$

After simplification,

$$r_0^2=\frac{k_1^2-g_1^2\Omega_{\mathrm{ZH}}^4\tau^2}{2\kappa^2}. \tag{35}$$

Therefore,

$$r_0 = \sqrt{\frac{k_1^2 - g_1^2\Omega_{\mathrm{ZH}}^4\tau^2}{2\kappa^2}}. \tag{36}$$

A nontrivial real zero exists precisely when

$$k_1^2 - g_1^2\Omega_{\mathrm{ZH}}^4\tau^2 > 0. \tag{37}$$

Under this condition, the averaged function has the zero

$$(r_0, u_0) = \left(\sqrt{\frac{k_1^2 - g_1^2\Omega_{\mathrm{ZH}}^4\tau^2}{2\kappa^2}},\ \frac{\omega_0^2}{2\kappa}\left(k_1 - g_1\Omega_{\mathrm{ZH}}^2\tau\right)\right). \tag{38}$$

2.7. **Nondegeneracy of the averaged zero.** The Jacobian matrix of the averaged function at (r_0, u_0) is

$$D\mathbf{f}_{\mathrm{ZH}}(r_0, u_0) = \begin{pmatrix} 0 & \dfrac{2\pi\kappa r_0}{\Omega_{\mathrm{ZH}}^3\tau\omega_0^2} \\ -\dfrac{2\pi\kappa\omega_0^2 r_0}{\Omega_{\mathrm{ZH}}^3\tau} & \dfrac{2\pi g_1}{\Omega_{\mathrm{ZH}}} \end{pmatrix}. \tag{39}$$

Its determinant is

$$\det D\mathbf{f}_{\mathrm{ZH}}(r_0, u_0) = \frac{2\pi^2\left(k_1^2 - g_1^2\Omega_{\mathrm{ZH}}^4\tau^2\right)}{\Omega_{\mathrm{ZH}}^6\tau^2}. \tag{40}$$

Condition (37) therefore implies $\det D\mathbf{f}_{\mathrm{ZH}}(r_0, u_0) > 0$. Thus the zero (r_0, u_0) is isolated and nondegenerate.

By the first-order averaging theorem, for all sufficiently small $|\varepsilon| > 0$, system (25) possesses a 2π-periodic solution satisfying

$$(r(\theta, \varepsilon), u(\theta, \varepsilon)) = (r_0, u_0) + O(\varepsilon). \tag{41}$$

This solution corresponds to a periodic orbit of the original oscillator–actuator system.

2.8. **Stability of the periodic orbit.** The trace and determinant of the averaged Jacobian are

$$\operatorname{tr} D\mathbf{f}_{\mathrm{ZH}}(r_0, u_0) = \frac{2\pi g_1}{\Omega_{\mathrm{ZH}}}, \tag{42}$$

and

$$\det D\mathbf{f}_{\mathrm{ZH}}(r_0, u_0) = \frac{2\pi^2\left(k_1^2 - g_1^2\Omega_{\mathrm{ZH}}^4\tau^2\right)}{\Omega_{\mathrm{ZH}}^6\tau^2} > 0. \tag{43}$$

The corresponding eigenvalues are

$$\lambda_{\pm}^{\mathrm{av}} = \frac{\pi g_1}{\Omega_{\mathrm{ZH}}} \pm \frac{\pi}{\Omega_{\mathrm{ZH}}^3\tau}\sqrt{3g_1^2\Omega_{\mathrm{ZH}}^4\tau^2 - 2k_1^2}. \tag{44}$$

Since the determinant is positive, the two eigenvalues have real parts with the same sign, determined by the trace. Therefore,

$$\operatorname{Re}\lambda_{\pm}^{\mathrm{av}} < 0 \quad \Longleftrightarrow \quad g_1 < 0,$$

and

$$\operatorname{Re}\lambda_{\pm}^{\mathrm{av}} > 0 \quad \Longleftrightarrow \quad g_1 > 0.$$

The nontrivial Floquet multipliers of the periodic orbit satisfy

$$\mu_j = 1 + \varepsilon\lambda_j^{\mathrm{av}} + O(\varepsilon^2), \qquad j = 1, 2. \tag{45}$$

Consequently, the periodic orbit is orbitally asymptotically stable when $\varepsilon g_1 < 0$ and unstable when $\varepsilon g_1 > 0$.

2.9. **Approximation of the bifurcating periodic orbit.** Returning to the original time variable, one has

$$\dot{\theta} = \Omega_{\mathrm{ZH}} + O(\varepsilon),$$

and hence

$$\theta(t,\varepsilon) = \Omega_{\mathrm{ZH}} t + \theta_0 + O(\varepsilon), \tag{46}$$

where θ_0 is an arbitrary initial phase.

Moreover,

$$r(t,\varepsilon) = r_0 + O(\varepsilon), \qquad u(t,\varepsilon) = u_0 + O(\varepsilon).$$

Therefore,

$$T_\varepsilon = \frac{2\pi}{\Omega_{\mathrm{ZH}}} + O(\varepsilon). \tag{47}$$

In the zero–Hopf coordinates,

$$\begin{aligned} X(t,\varepsilon) &= \varepsilon r_0 \cos\left(\Omega_{\mathrm{ZH}} t + \theta_0\right) + O(\varepsilon^2), \\ Y(t,\varepsilon) &= \varepsilon r_0 \sin\left(\Omega_{\mathrm{ZH}} t + \theta_0\right) + O(\varepsilon^2), \\ Z(t,\varepsilon) &= \varepsilon u_0 + O(\varepsilon^2). \end{aligned} \tag{48}$$

Using (21), the periodic orbit in the original variables is

$$\begin{aligned} x(t,\varepsilon) &= \varepsilon \left[r_0 \cos\left(\Omega_{\mathrm{ZH}} t + \theta_0\right) + \frac{u_0}{\omega_0^2} \right] + O(\varepsilon^2), \\ v(t,\varepsilon) &= -\varepsilon \Omega_{\mathrm{ZH}} r_0 \sin\left(\Omega_{\mathrm{ZH}} t + \theta_0\right) + O(\varepsilon^2), \\ f(t,\varepsilon) &= \varepsilon \left[\frac{r_0}{\tau^2} \cos\left(\Omega_{\mathrm{ZH}} t + \theta_0\right) + \frac{\Omega_{\mathrm{ZH}} r_0}{\tau} \sin\left(\Omega_{\mathrm{ZH}} t + \theta_0\right) + u_0 \right] + O(\varepsilon^2). \end{aligned} \tag{49}$$

Substituting (34) and (36), one obtains

$$\begin{aligned} x(t,\varepsilon) &= \varepsilon \sqrt{\frac{k_1^2 - g_1^2 \Omega_{\mathrm{ZH}}^4 \tau^2}{2\kappa^2}} \cos\left(\Omega_{\mathrm{ZH}} t + \theta_0\right) \\ &\quad + \frac{\varepsilon}{2\kappa}\left(k_1 - g_1 \Omega_{\mathrm{ZH}}^2 \tau\right) + O(\varepsilon^2), \\ v(t,\varepsilon) &= -\varepsilon \Omega_{\mathrm{ZH}} \sqrt{\frac{k_1^2 - g_1^2 \Omega_{\mathrm{ZH}}^4 \tau^2}{2\kappa^2}} \sin\left(\Omega_{\mathrm{ZH}} t + \theta_0\right) + O(\varepsilon^2), \\ f(t,\varepsilon) &= \frac{\varepsilon}{\tau^2} \sqrt{\frac{k_1^2 - g_1^2 \Omega_{\mathrm{ZH}}^4 \tau^2}{2\kappa^2}} \cos\left(\Omega_{\mathrm{ZH}} t + \theta_0\right) \\ &\quad + \frac{\varepsilon \Omega_{\mathrm{ZH}}}{\tau} \sqrt{\frac{k_1^2 - g_1^2 \Omega_{\mathrm{ZH}}^4 \tau^2}{2\kappa^2}} \sin\left(\Omega_{\mathrm{ZH}} t + \theta_0\right) \\ &\quad + \frac{\varepsilon \omega_0^2}{2\kappa}\left(k_1 - g_1 \Omega_{\mathrm{ZH}}^2 \tau\right) + O(\varepsilon^2). \end{aligned} \tag{50}$$

Therefore, whenever $k_1^2 - g_1^2 \Omega_{\mathrm{ZH}}^4 \tau^2 > 0$, a periodic orbit bifurcates from the zero–Hopf equilibrium. Its amplitude is proportional to $|\varepsilon|$, its leading angular frequency is Ω_{ZH}, and its leading period is $2\pi/\Omega_{\mathrm{ZH}}$. Its orbital stability is determined by the sign of εg_1.

3. Classical Hopf bifurcation

We investigate the classical Hopf bifurcation of the equilibrium $E_0 = (0, 0, 0)$ of system (1). Throughout this section, we assume

$$\omega_0 > 0, \qquad \tau > 0, \qquad \kappa \neq 0, \qquad \omega_0 \tau > 1.$$

In contrast with the zero–Hopf configuration considered previously, the classical Hopf configuration consists of a simple pair of nonzero purely imaginary eigenvalues and a nonzero real eigenvalue. We first determine the spectral locus and verify transversality. We then compute the nonlinear bifurcation coefficient through second-order averaging and establish existence, local uniqueness, and stability of the bifurcating periodic orbit.

The characteristic polynomial of the linearization at E_0 is

$$p(\lambda; g, k_s) = \lambda^3 + a_1(g)\lambda^2 + a_2(g)\lambda + a_3(k_s), \tag{51}$$

where

$$a_1(g) = d - g + \frac{1}{\tau}, \qquad a_2(g) = \omega_0^2 + \frac{d-g}{\tau}, \qquad a_3(k_s) = \frac{\omega_0^2 - k_s}{\tau}. \tag{52}$$

Suppose that p has roots $\pm i\varpi$, with $\varpi > 0$. Substituting $\lambda = i\varpi$ into (51) and separating real and imaginary parts gives

$$a_3 - a_1\varpi^2 = 0, \qquad \varpi(a_2 - \varpi^2) = 0.$$

Consequently,

$$\varpi^2 = a_2 > 0, \qquad a_3 = a_1 a_2.$$

Requiring the remaining real eigenvalue to be nonzero yields

$$\frac{\omega_0^2 - k_s}{\tau} = \left(d - g + \frac{1}{\tau}\right)\left(\omega_0^2 + \frac{d-g}{\tau}\right), \tag{53}$$

subject to

$$\omega_0^2 + \frac{d-g}{\tau} > 0, \qquad d - g + \frac{1}{\tau} \neq 0. \tag{54}$$

Under these conditions,

$$p(\lambda; g, k_s) = \left(\lambda + d - g + \frac{1}{\tau}\right)\left(\lambda^2 + \omega_0^2 + \frac{d-g}{\tau}\right). \tag{55}$$

These relations determine the spectral Hopf locus. Nonlinear nondegeneracy will be established below.

Introduce $\sigma = d - g + \frac{1}{\tau}$ and define $g_{\mathrm{H}}(\sigma) = d + \frac{1}{\tau} - \sigma$. The corresponding oscillatory frequency is

$$\Omega_{\mathrm{H}}(\sigma) = \sqrt{\omega_0^2 - \frac{1}{\tau^2} + \frac{\sigma}{\tau}} = \sqrt{\Omega_{\mathrm{ZH}}^2 + \frac{\sigma}{\tau}}. \tag{56}$$

Its domain is

$$\mathcal{I}_{\mathrm{H}} = \left\{\sigma \in \mathbb{R} : \Omega_{\mathrm{H}}^2(\sigma) > 0\right\} = \left(-\tau\Omega_{\mathrm{ZH}}^2, \infty\right). \tag{57}$$

For $\sigma \in \mathcal{I}_{\mathrm{H}}$, set

$$k_{s,\mathrm{H}}(\sigma) = \omega_0^2 - \tau\sigma\Omega_{\mathrm{H}}^2(\sigma). \tag{58}$$

At the following parameter values the coefficients in (52) satisfy

$$a_1 = \sigma, \qquad a_2 = \Omega_{\mathrm{H}}^2(\sigma), \qquad a_3 = \sigma\Omega_{\mathrm{H}}^2(\sigma).$$

Substitution into (51) gives the stated factorization. Thus we have this proposition:

Proposition 3.1. *At $g = g_{\mathrm{H}}(\sigma), k_s = k_{s,\mathrm{H}}(\sigma)$ and $\sigma \in \mathcal{I}_{\mathrm{H}}$, the characteristic polynomial at E_0 factors as*

$$p_{\mathrm{H}}(\lambda; \sigma) = (\lambda + \sigma)\left(\lambda^2 + \Omega_{\mathrm{H}}^2(\sigma)\right).$$

Hence its eigenvalues are $-\sigma, \pm i\Omega_{\mathrm{H}}(\sigma)$.

For $\sigma > 0$, the noncritical direction is stable, whereas for $\sigma < 0$ it is unstable. At $\sigma = 0$,

$$g_{\mathrm{H}}(0) = d + \frac{1}{\tau}, \qquad k_{s,\mathrm{H}}(0) = \omega_0^2, \qquad \Omega_{\mathrm{H}}(0) = \Omega_{\mathrm{ZH}},$$

and the spectrum becomes $0, \pm i\Omega_{\mathrm{ZH}}$. Thus the zero–Hopf point is the limiting spectral configuration on this parametrized curve.

Fix $\sigma \in \mathcal{I}_{\mathrm{H}} \setminus \{0\}$and $g = g_{\mathrm{H}}(\sigma)$, and use k_s as the bifurcation parameter. For the remainder of the calculations, write

$$\Omega = \Omega_{\mathrm{H}}(\sigma), \qquad D_\sigma = \Omega^2 + \sigma^2 > 0. \tag{59}$$

Let $\lambda(k_s)$ be the eigenvalue branch satisfying $\lambda(k_{s,\mathrm{H}}(\sigma)) = i\Omega$. Since this eigenvalue is simple, implicit differentiation gives $\lambda'(k_s) = -\frac{p_{k_s}}{p_\lambda} = \frac{1}{\tau p_\lambda}$. At the critical parameter,

$$p_\lambda(i\Omega) = -2\Omega^2 + 2i\sigma\Omega.$$

Therefore,

$$\frac{d}{dk_s} \operatorname{Re} \lambda(k_s)\Big|_{k_s = k_{s,\mathrm{H}}(\sigma)} = -\frac{1}{2\tau D_\sigma} =: \beta_{\mathrm{H}} < 0. \tag{60}$$

In particular, the spectral crossing is transversal. With $\mu = k_s - k_{s,\mathrm{H}}(\sigma)$, the critical pair satisfies

$$\operatorname{Re} \lambda = -\frac{\mu}{2\tau D_\sigma} + O(\mu^2).$$

Thus the pair lies in the right half-plane for $\mu < 0$ and in the left half-plane for $\mu > 0$, provided $|\mu|$ is sufficiently small.

We consider the unfolding

$$g = g_{\mathrm{H}}(\sigma), \qquad k_s = k_{s,\mathrm{H}}(\sigma) + \eta_2\varepsilon^2 + \eta_3\varepsilon^3, \tag{61}$$

where $\eta_2, \eta_3 \in \mathbb{R}$ are fixed. The scaling $\mu = O(\varepsilon^2)$ is adapted to periodic orbits of amplitude $O(|\varepsilon|)$.

The critical spectral configuration occurs at $\varepsilon = 0$. For nonzero ε, the equilibrium generally lies off the Hopf locus. Indeed, under (61),

$$p(\lambda) = (\lambda + \sigma)(\lambda^2 + \Omega^2) - \frac{\eta_2\varepsilon^2 + \eta_3\varepsilon^3}{\tau}.$$

A nonzero purely imaginary root requires $\eta_2\varepsilon^2 + \eta_3\varepsilon^3 = 0$. In particular, if $\eta_2 \neq 0$, no such root exists for sufficiently small nonzero ε.

All estimates below are understood for fixed $\sigma \neq 0$ and fixed remaining parameters. Their constants and the admissible size of $|\varepsilon|$ may depend on these parameters.

3.1. **The first averaged function and transverse correction.** At $\varepsilon = 0$, the Jacobian at E_0 is

$$A_\sigma = \begin{pmatrix} 0 & 1 & 0 \\ -\omega_0^2 & \frac{1}{\tau} - \sigma & 1 \\ \dfrac{k_{s,\mathrm{H}}(\sigma)}{\tau} & 0 & -\dfrac{1}{\tau} \end{pmatrix}.$$

Define

$$a_\sigma = \frac{1 - \sigma\tau}{\tau^2}, \qquad b_\sigma = \frac{\Omega(1 - \sigma\tau)}{\tau}, \qquad c_\sigma = \omega_0^2 + \frac{\sigma}{\tau},$$

and

$$P_\sigma = \begin{pmatrix} 1 & 0 & 1 \\ 0 & -\Omega & -\sigma \\ a_\sigma & b_\sigma & c_\sigma \end{pmatrix}.$$

Its columns satisfy

$$A_\sigma p_1 = \Omega p_2, \qquad A_\sigma p_2 = -\Omega p_1, \qquad A_\sigma p_3 = -\sigma p_3.$$

Moreover, $\det P_\sigma = -\Omega D_\sigma \neq 0$. Consequently,

$$P_\sigma^{-1} A_\sigma P_\sigma = \begin{pmatrix} 0 & -\Omega & 0 \\ \Omega & 0 & 0 \\ 0 & 0 & -\sigma \end{pmatrix}.$$

Introduce (X, Y, Z) by

$$\begin{pmatrix} x \\ v \\ f \end{pmatrix} = P_\sigma \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}.$$

Explicitly,

$$x = X + Z, \qquad v = -\Omega Y - \sigma Z, \qquad f = a_\sigma X + b_\sigma Y + c_\sigma Z.$$

Set

$$\mathcal{H}(X, Z, \varepsilon) = \frac{\eta_2 \varepsilon^2 + \eta_3 \varepsilon^3}{\tau}(X + Z) - \frac{\kappa}{\tau}(X + Z)^2. \tag{62}$$

Since

$$P_\sigma^{-1} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1/D_\sigma \\ -\sigma/(\Omega D_\sigma) \\ 1/D_\sigma \end{pmatrix},$$

the full transformed system is

$$\begin{aligned} \dot{X} &= -\Omega Y - \frac{\mathcal{H}}{D_\sigma}, \\ \dot{Y} &= \Omega X - \frac{\sigma \mathcal{H}}{\Omega D_\sigma}, \\ \dot{Z} &= -\sigma Z + \frac{\mathcal{H}}{D_\sigma}. \end{aligned}$$

For $\varepsilon \neq 0$, introduce

$$X = \varepsilon R \cos\theta, \qquad Y = \varepsilon R \sin\theta, \qquad Z = \varepsilon W. \tag{63}$$

The analysis is carried out locally with R in a compact subinterval of $(0, \infty)$ and W near zero. Thus the division by R below causes no singularity.

Define

$$S(\theta, R, W) = R\cos\theta + W$$

and

$$A(\theta) = \frac{\Omega \cos\theta + \sigma \sin\theta}{\Omega D_\sigma}, \qquad B(\theta) = \frac{\Omega \sin\theta - \sigma \cos\theta}{\Omega D_\sigma}.$$

Then

$$\mathcal{H} = -\frac{\kappa \varepsilon^2}{\tau} S^2 + \frac{\eta_2 \varepsilon^3}{\tau} S + \frac{\eta_3 \varepsilon^4}{\tau} S.$$

Thus we have

$$\begin{aligned} \dot{R} &= \frac{\kappa \varepsilon}{\tau} S^2 A - \frac{\eta_2 \varepsilon^2}{\tau} SA + O(|\varepsilon|^3), \\ \dot{\theta} &= \Omega - \frac{\kappa \varepsilon}{\tau R} S^2 B + \frac{\eta_2 \varepsilon^2}{\tau R} SB + O(|\varepsilon|^3), \\ \dot{W} &= -\sigma W - \frac{\kappa \varepsilon}{\tau D_\sigma} S^2 + \frac{\eta_2 \varepsilon^2}{\tau D_\sigma} S + O(|\varepsilon|^3). \end{aligned}$$

The remainders are smooth and uniform on the compact neighborhoods under consideration.

Since $\dot{\theta} = \Omega + O(|\varepsilon|)$ and $\Omega > 0$, we have $\dot{\theta} > 0$ for sufficiently small $|\varepsilon|$. We may therefore use θ as the independent variable. Write

$$q_1 = -\frac{\kappa}{\tau R} S^2 B, \qquad q_2 = \frac{\eta_2}{\tau R} SB.$$

Then

$$\frac{1}{\dot\theta} = \frac{1}{\Omega} - \varepsilon\frac{q_1}{\Omega^2} + \varepsilon^2\left(\frac{q_1^2}{\Omega^3} - \frac{q_2}{\Omega^2}\right) + O(|\varepsilon|^3).$$

With a prime denoting differentiation with respect to θ, the resulting system is

$$\begin{pmatrix} R \\ W \end{pmatrix}' = F_0(\theta, R, W) + \varepsilon F_1(\theta, R, W) + \varepsilon^2 F_2(\theta, R, W) + O(|\varepsilon|^3),$$

where

$$F_0 = \begin{pmatrix} 0 \\ -\sigma W/\Omega \end{pmatrix},$$

and

$$\begin{aligned} F_{1,1} &= \frac{\kappa S^2}{\tau\Omega} A, \\ F_{1,2} &= -\frac{\kappa S^2}{\tau\Omega D_\sigma} - \frac{\sigma\kappa W S^2}{\tau R\Omega^2} B. \end{aligned}$$

The second-order terms are

$$\begin{aligned} F_{2,1} &= -\frac{\eta_2 S}{\tau\Omega} A + \frac{\kappa^2 S^4}{\tau^2 R\Omega^2} AB, \\ F_{2,2} &= \frac{\eta_2 S}{\tau\Omega D_\sigma} - \frac{\kappa^2 S^4}{\tau^2 D_\sigma R\Omega^2} B \\ &\quad - \frac{\sigma\kappa^2 W S^4}{\tau^2 R^2\Omega^3} B^2 + \frac{\sigma\eta_2 W S}{\tau R\Omega^2} B. \end{aligned}$$

All these functions are 2π-periodic in θ.

At $\varepsilon = 0$, system reduces to

$$R' = 0, \qquad W' = -\frac{\sigma}{\Omega} W.$$

Its solutions are

$$R(\theta) = R_0, \qquad W(\theta) = W_0 e^{-\sigma\theta/\Omega}.$$

Thus $W_0 = 0$ gives a one-dimensional family of 2π-periodic solutions. The transverse return multiplier is $m_\sigma = e^{-2\pi\sigma/\Omega} \neq 1$. This nondegeneracy allows us to solve the transverse periodicity condition locally.

On $W = 0$,

$$F_{1,1}(\theta, R_0, 0) = \frac{\kappa R_0^2 \cos^2\theta}{\tau\Omega^2 D_\sigma} \left(\Omega\cos\theta + \sigma\sin\theta\right).$$

Hence the first averaged radial function is

$$f_{\mathrm{H},1}(R_0) = \int_0^{2\pi} F_{1,1}(\theta, R_0, 0)\, d\theta = 0. \tag{64}$$

To compute the next radial contribution, fix the initial radial coordinate $R(0) = R_0$ and write

$$\begin{aligned} R(\theta) &= R_0 + \varepsilon r_1(\theta; R_0) + O(\varepsilon^2), \\ W(\theta) &= \varepsilon w_1(\theta; R_0) + O(\varepsilon^2). \end{aligned}$$

We impose $r_1(0; R_0) = 0$ and choose w_1 to be 2π-periodic.

The radial correction satisfies $r_1' = F_{1,1}(\theta, R_0, 0)$. Integration gives

$$r_1(\theta; R_0) = \frac{\kappa R_0^2}{\tau\Omega^2 D_\sigma}\left[\Omega\left(\sin\theta - \frac{\sin^3\theta}{3}\right) + \frac{\sigma}{3}(1 - \cos^3\theta)\right].$$

In particular, $r_1(2\pi; R_0) = r_1(0; R_0) = 0$. The transverse correction satisfies

$$w_1' + \frac{\sigma}{\Omega} w_1 = -\frac{\kappa R_0^2}{\tau\Omega D_\sigma}\cos^2\theta.$$

Since $\sigma \neq 0$, its homogeneous equation has no nonzero 2π-periodic solution. Therefore its periodic solution is unique. Using $\cos^2\theta = (1+\cos 2\theta)/2$, we obtain

$$\begin{aligned} w_1(\theta;R_0) = &-\frac{\kappa R_0^2}{2\tau\sigma D_\sigma} \\ &-\frac{\kappa R_0^2\sigma}{2\tau D_\sigma(4\Omega^2+\sigma^2)}\cos 2\theta \\ &-\frac{\kappa R_0^2\Omega}{\tau D_\sigma(4\Omega^2+\sigma^2)}\sin 2\theta. \end{aligned}$$

Define

$$\gamma_\sigma(R_0) = w_1(0;R_0) = -\frac{\kappa R_0^2(\sigma^2+2\Omega^2)}{\tau\sigma D_\sigma(\sigma^2+4\Omega^2)}.$$

The transverse periodicity condition therefore requires $W(0) = \varepsilon\gamma_\sigma(R_0) + O(\varepsilon^2)$.

3.2. **The second-order radial bifurcation function.** The second-order radial displacement, after imposing transverse periodicity, is determined by

$$\begin{aligned} f_{\mathrm{H},2}(R_0) = \int_0^{2\pi} \Big[&F_{2,1}(\theta,R_0,0) + \partial_R F_{1,1}(\theta,R_0,0) r_1(\theta;R_0) \\ &+ \partial_W F_{1,1}(\theta,R_0,0) w_1(\theta;R_0)\Big]\, d\theta. \end{aligned}$$

We evaluate these three contributions separately. First,

$$F_{2,1}(\theta,R_0,0) = -\frac{\eta_2 R_0\cos\theta}{\tau\Omega}A(\theta) + \frac{\kappa^2 R_0^3\cos^4\theta}{\tau^2\Omega^2}A(\theta)B(\theta),$$

and so

$$\int_0^{2\pi} F_{2,1}(\theta,R_0,0)\, d\theta = -\frac{\pi\eta_2 R_0}{\tau\Omega D_\sigma} - \frac{\pi\kappa^2\sigma R_0^3}{2\tau^2\Omega^3 D_\sigma^2}.$$

Second, since

$$\partial_R F_{1,1}(\theta,R_0,0) = \frac{2}{R_0}F_{1,1}(\theta,R_0,0) = \frac{2}{R_0}r_1'(\theta;R_0),$$

we have

$$\begin{aligned} \int_0^{2\pi} \partial_R F_{1,1}(\theta,R_0,0) r_1(\theta;R_0)\, d\theta &= \frac{1}{R_0}\left[r_1(\theta;R_0)^2\right]_0^{2\pi} \\ &= 0. \end{aligned}$$

Third,

$$\partial_W F_{1,1}(\theta,R_0,0) = \frac{2\kappa R_0\cos\theta}{\tau\Omega^2 D_\sigma}\left(\Omega\cos\theta + \sigma\sin\theta\right).$$

we obtain

$$\begin{aligned} &\int_0^{2\pi} \partial_W F_{1,1}(\theta,R_0,0) w_1(\theta;R_0)\, d\theta \\ &\qquad = -\frac{\pi\kappa^2 R_0^3}{\tau^2\Omega\sigma D_\sigma^2} - \frac{3\pi\kappa^2\sigma R_0^3}{2\tau^2\Omega D_\sigma^2(4\Omega^2+\sigma^2)}. \end{aligned}$$

Combining these expressions gives

$$\begin{aligned} f_{\mathrm{H},2}(R_0) = &-\frac{\pi\eta_2}{\tau\Omega D_\sigma}R_0 \\ &-\frac{\pi\kappa^2(8\Omega^2+\sigma^2)}{2\tau^2\Omega^3\sigma D_\sigma(4\Omega^2+\sigma^2)}R_0^3. \end{aligned}$$

In simplifying the cubic term, we use

$$\sigma^4 + 9\Omega^2\sigma^2 + 8\Omega^4 = (\Omega^2+\sigma^2)(8\Omega^2+\sigma^2).$$

Equivalently,

$$f_{\mathrm{H},2}(R_0) = -\frac{\pi R_0}{\tau\Omega D_\sigma}\left[\eta_2 + \mathcal{C}_{\mathrm{H}}(\sigma)R_0^2\right], \tag{65}$$

where

$$\mathcal{C}_{\mathrm{H}}(\sigma) = \frac{\kappa^2(8\Omega^2+\sigma^2)}{2\tau\Omega^2\sigma(4\Omega^2+\sigma^2)}.$$

A positive zero exists precisely when $\sigma\eta_2 < 0$. Under this condition, it is

$$R_{\mathrm{H}}^* = \sqrt{-\frac{2\tau\Omega^2\sigma(4\Omega^2+\sigma^2)}{\kappa^2(8\Omega^2+\sigma^2)}\,\eta_2}. \tag{66}$$

Differentiating (65), and using $\mathcal{C}_{\mathrm{H}}(\sigma)(R_{\mathrm{H}}^*)^2 = -\eta_2$, gives

$$f'_{\mathrm{H},2}(R_{\mathrm{H}}^*) = \frac{2\pi\eta_2}{\tau\Omega D_\sigma} \neq 0. \tag{67}$$

Thus the positive zero is simple.

3.3. **Existence and local uniqueness.** We give the return-map argument underlying the preceding averaging calculation. This also specifies the local uniqueness statement.

Assume $\sigma\eta_2 < 0$ and choose a compact interval $J \subset (0,\infty)$ whose interior contains R_{H}^*. For initial conditions near $J \times \{0\}$, let

$$\mathcal{P}_\varepsilon(R_0, W_0) = \big(\mathcal{P}_\varepsilon^R(R_0, W_0), \mathcal{P}_\varepsilon^W(R_0, W_0)\big)$$

be the return map from $\theta = 0$ to $\theta = 2\pi$. The map is smooth in its arguments and in ε. At $\varepsilon = 0$,

$$\mathcal{P}_0(R_0, W_0) = (R_0, m_\sigma W_0), \qquad m_\sigma = e^{-2\pi\sigma/\Omega}.$$

Since $m_\sigma - 1 \neq 0$, the implicit function theorem applied to $\mathcal{P}_\varepsilon^W(R_0, W_0) - W_0 = 0$ gives a unique smooth function $W_0 = \psi(R_0, \varepsilon)$ near $J \times \{0\}$, with

$$\psi(R_0, \varepsilon) = \varepsilon\gamma_\sigma(R_0) + O(\varepsilon^2). \tag{68}$$

Define the reduced radial displacement $\Phi(R_0, \varepsilon) = \mathcal{P}_\varepsilon^R\big(R_0, \psi(R_0, \varepsilon)\big) - R_0$. The first-order term vanishes by (64) and

$$\Phi(R_0, \varepsilon) = \varepsilon^2 f_{\mathrm{H},2}(R_0) + O(|\varepsilon|^3),$$

with the corresponding expansion also valid after differentiation with respect to R_0.

The quotient $\Phi(R_0, \varepsilon)/\varepsilon^2$ extends smoothly to $\varepsilon = 0$, where it equals $f_{\mathrm{H},2}(R_0)$. By (67), a second application of the implicit function theorem gives a unique root

$$R_0 = R_\varepsilon = R_{\mathrm{H}}^* + O(|\varepsilon|)$$

near R_{H}^*. Consequently, $(R_\varepsilon, \psi(R_\varepsilon, \varepsilon))$ is a fixed point of $\mathcal{P}_\varepsilon$ for every sufficiently small nonzero ε.

Since $R_\varepsilon > 0$ and $\dot\theta > 0$, this fixed point represents a nonconstant periodic orbit of the original system. Its initial scaled coordinates satisfy

$$R(0, \varepsilon) = R_{\mathrm{H}}^* + O(|\varepsilon|),$$
$$W(0, \varepsilon) = \varepsilon\gamma_\sigma(R_{\mathrm{H}}^*) + O(\varepsilon^2).$$

The return-map argument proves uniqueness near this positive scaled radius.

To establish the usual local Hopf uniqueness statement, consider the radial normal form on a local center manifold. Normalize its amplitude ρ so that

$$\rho = \sqrt{X^2 + Y^2} + O(X^2 + Y^2)$$

at the critical parameter. To the required order, the radial equation is

$$\dot\rho = \beta_{\mathrm{H}}\mu\rho + \ell_{\mathrm{H}}(\sigma)\rho^3 + O\left(\mu^2\rho + |\mu|\rho^3 + \rho^5\right).$$

The cubic coefficient is obtained from the cubic return displacement by multiplying by $\Omega/(2\pi)$:

$$\ell_{\mathrm{H}}(\sigma) = -\frac{\kappa^2(8\Omega^2+\sigma^2)}{4\tau^2\Omega^2\sigma(\Omega^2+\sigma^2)(4\Omega^2+\sigma^2)}.$$

This expression refers to the stated amplitude normalization and to the original time variable.

Since $\beta_{\mathrm{H}} = -\frac{1}{2\tau D_\sigma} \neq 0$ and $\ell_{\mathrm{H}}(\sigma) \neq 0$, the Hopf bifurcation is nondegenerate. The nondegenerate Hopf theorem therefore gives local uniqueness of the bifurcating periodic orbit, up to time translation, in a sufficiently small neighborhood of E_0. This neighborhood is chosen for the fixed value of σ and need not remain uniform as $\sigma \to 0$.

Finally,

$$\operatorname{sgn} \ell_{\mathrm{H}}(\sigma) = -\operatorname{sgn}\sigma.$$

Thus the bifurcation is supercritical for $\sigma > 0$ and subcritical for $\sigma < 0$. Here criticality is understood in the usual Hopf sense, with the real part of the critical eigenvalues as the unfolding coordinate.

3.4. **Approximation of the periodic orbit.** Assume $\sigma\eta_2 < 0$. The periodic solution constructed above satisfies

$$R(\theta,\varepsilon) = R^*_{\mathrm{H}} + O(|\varepsilon|), \qquad W(\theta,\varepsilon) = O(|\varepsilon|), \tag{69}$$

uniformly in θ over one period. Periodicity extends these bounds to all θ.

More precisely,

$$W(\theta,\varepsilon) = \varepsilon w_1(\theta; R^*_{\mathrm{H}}) + O(\varepsilon^2).$$

Consequently,

$$Z(\theta,\varepsilon) = \varepsilon^2 w_1(\theta; R^*_{\mathrm{H}}) + O(|\varepsilon|^3).$$

Since $\dot\theta = \Omega + O(|\varepsilon|)$,

$$\theta(t,\varepsilon) = \Omega t + \theta_0 + O(|\varepsilon|),$$

uniformly for t in bounded intervals. Returning to the original variables, we obtain

$$\begin{aligned} x(t,\varepsilon) &= \varepsilon R^*_{\mathrm{H}}\cos(\Omega t+\theta_0) + O(\varepsilon^2),\\ v(t,\varepsilon) &= -\varepsilon\Omega R^*_{\mathrm{H}}\sin(\Omega t+\theta_0) + O(\varepsilon^2),\\ f(t,\varepsilon) &= \varepsilon\frac{1-\sigma\tau}{\tau^2}R^*_{\mathrm{H}}\cos(\Omega t+\theta_0)\\ &\quad + \varepsilon\frac{\Omega(1-\sigma\tau)}{\tau}R^*_{\mathrm{H}}\sin(\Omega t+\theta_0) + O(\varepsilon^2). \end{aligned} \tag{70}$$

All remainders in (70) are uniform on bounded time intervals. No uniform-in-time claim is made with the limiting frequency Ω.

The period is

$$T_\varepsilon = \int_0^{2\pi} \frac{d\theta}{\dot\theta(\theta, R(\theta,\varepsilon), W(\theta,\varepsilon),\varepsilon)}.$$

On the leading periodic solution,

$$q_1(\theta, R^*_{\mathrm{H}}, 0) = -\frac{\kappa R^*_{\mathrm{H}}}{\tau}\cos^2\theta\, B(\theta).$$

Its integral over a period vanishes, because both $\cos^2\theta\sin\theta$ and $\cos^3\theta$ have zero integral. Therefore,

$$T_\varepsilon = \frac{2\pi}{\Omega} + O(\varepsilon^2). \tag{71}$$

In particular, the displacement amplitude is $|\varepsilon|R^*_{\mathrm{H}} + O(\varepsilon^2)$, and the leading oscillatory frequency is Ω.

3.5. **Floquet exponents and orbital stability.** At $\varepsilon = 0$, the return-map multipliers are 1 and $m_\sigma = e^{-2\pi\sigma/\Omega}$. They are distinct because $\sigma \neq 0$. Thus the multiplier near 1 can be followed smoothly. The reduced displacement calculation gives

$$\mu_{\mathrm{rad}} = 1 + \varepsilon^2 f'_{\mathrm{H},2}(R^*_{\mathrm{H}}) + O(|\varepsilon|^3).$$

endequation Here the elimination of the transverse fixed-point equation gives the same leading correction as the eigenvalue near 1 of the full return-map derivative: the other unperturbed multiplier is separated from 1.

Using (67) and (71), the radial Floquet exponent in the original time variable is

$$\begin{aligned}\Lambda_{\mathrm{rad}} &= \frac{1}{T_\varepsilon}\log\mu_{\mathrm{rad}} \\ &= \frac{\eta_2}{\tau D_\sigma}\varepsilon^2 + O(|\varepsilon|^3).\end{aligned}$$

Hence the radial direction is attracting for $\eta_2 < 0$ and repelling for $\eta_2 > 0$.

An exact divergence identity determines the transverse exponent more precisely. The divergence of the original vector field is $\operatorname{div} F = g - d - \frac{1}{\tau}$. Along the unfolding (61), this quantity is identically $-\sigma$. Liouville's formula, together with the trivial Floquet multiplier 1 of an autonomous periodic orbit, gives

$$\mu_{\mathrm{rad}}\mu_{\mathrm{tr}} = e^{-\sigma T_\varepsilon}. \tag{72}$$

Both nontrivial multipliers are positive for sufficiently small $|\varepsilon|$, since their limits are 1 and $m_\sigma > 0$. Taking real logarithms therefore yields

$$\Lambda_{\mathrm{rad}} + \Lambda_{\mathrm{tr}} = -\sigma. \tag{73}$$

Consequently,

$$\Lambda_{\mathrm{tr}} = -\sigma - \frac{\eta_2}{\tau D_\sigma}\varepsilon^2 + O(|\varepsilon|^3). \tag{74}$$

If $\sigma > 0$, the existence condition requires $\eta_2 < 0$. Both nontrivial exponents are then negative, and the periodic orbit is orbitally asymptotically stable.

If $\sigma < 0$, the existence condition requires $\eta_2 > 0$. Both nontrivial exponents are then positive, and the periodic orbit is repelling in the two directions transverse to the flow.

Remark 3.2 (Lack of uniformity near the zero–Hopf point). The preceding analysis is performed for fixed $\sigma \neq 0$. As $\sigma \to 0$,

$$m_\sigma - 1 = e^{-2\pi\sigma/\Omega} - 1 \longrightarrow 0,$$

so the transverse periodicity equation loses its nondegeneracy. The factor $1/\sigma$ in γ_σ and in the cubic coefficient reflects the same loss of a noncritical transverse direction. Accordingly, neither the neighborhood of local uniqueness nor the bounds on $|\varepsilon|$ are asserted to be uniform as $\sigma \to 0$. At $\sigma = 0$, the two-dimensional zero–Hopf averaged system must be used.

The spectral factorization follows from Proposition 3.1. Equation (60) establishes transversality. The second-order calculation gives the positive simple zero (66) when $\sigma\eta_2 < 0$.

The return-map gives a periodic orbit for every sufficiently small nonzero ε. The nonvanishing crossing and cubic coefficients establish a nondegenerate Hopf bifurcation and its local uniqueness up to time translation.

The coordinate transformation and the estimates in Subsection 3.4 give the stated component expansions and period. Finally, (73) gives the Floquet exponents.

4. Dynamics at infinity

We study system (1) on the sphere at infinity using the Poincaré compactification. We first examine the equations in the local charts and then use these descriptions to determine the connections between the equilibria on the sphere. Throughout this section, assume $\omega_0 > 0$, $\tau > 0$, and $\kappa \neq 0$, and write

$$a = \omega_0^2, \qquad b = d - g, \qquad c = \frac{k_s}{\tau}, \qquad h = \frac{1}{\tau}, \qquad k = \frac{\kappa}{\tau}.$$

Then the system becomes

$$\dot{x} = v, \qquad \dot{v} = -ax - bv + f, \qquad \dot{f} = cx - hf - kx^2.$$

Since $k \neq 0$, this vector field has degree two.

4.1. **Compactification in the local charts.** Let U_1, U_2, U_3 be the charts corresponding to the positive x-, v-, and f-directions, and let V_1, V_2, V_3 be their negative counterparts. In each chart, infinity is represented by $w = 0$, while $w > 0$ corresponds to finite points. We use the time change $\frac{dt}{ds} = w$.

The resulting polynomial equations represent the compactified flow up to a positive smooth time factor in each chart.

In U_1, set

$$u = \frac{v}{x}, \qquad q = \frac{f}{x}, \qquad w = \frac{1}{x}.$$

The quotient rule gives

$$\frac{du}{dt} = -a - bu + q - u^2, \qquad \frac{dq}{dt} = c - hq - uq - \frac{k}{w}, \qquad \frac{dw}{dt} = -uw.$$

After the time change, we obtain

$$\begin{aligned} u' &= w(-a - bu + q - u^2), \\ q' &= -k + w(c - hq - uq), \\ w' &= -uw^2. \end{aligned} \tag{75}$$

On $w = 0$, this reduces to

$$u' = 0, \qquad q' = -k. \tag{76}$$

Thus there are no equilibria at infinity in U_1.

In U_2, take

$$u = \frac{x}{v}, \qquad q = \frac{f}{v}, \qquad w = \frac{1}{v}, \qquad L(u, q) = -au - b + q.$$

The same calculation yields

$$\begin{aligned} u' &= w\big(1 - uL(u, q)\big), \\ q' &= -ku^2 + w\big(cu - hq - qL(u, q)\big), \\ w' &= -w^2 L(u, q). \end{aligned} \tag{77}$$

Hence the equations at infinity are

$$u' = 0, \qquad q' = -ku^2. \tag{78}$$

Their equilibria form the line $u = 0$.

Finally, in U_3, set

$$u = \frac{x}{f}, \qquad q = \frac{v}{f}, \qquad w = \frac{1}{f}.$$

The compactified equations are

$$\begin{aligned} u' &= ku^3 + w(q - cu^2 + hu), \\ q' &= ku^2 q + w(-au - bq + 1 - cuq + hq), \\ w' &= ku^2 w - w^2(cu - h). \end{aligned} \tag{79}$$

Their restriction to $w = 0$ is

$$u' = ku^3, \qquad q' = ku^2 q, \tag{80}$$

whose equilibrium set is again the line $u = 0$.

For the negative charts, use

$$V_1 : (x, v, f) = \frac{(-1, -u, -q)}{w}, \qquad V_2 : (x, v, f) = \frac{(-u, -1, -q)}{w}, \qquad V_3 : (x, v, f) = \frac{(-u, -q, -1)}{w},$$

with $w > 0$ and the same time change. On $w = 0$, we find

$$\begin{aligned} V_1 : &\quad u' = 0, \qquad q' = k, \\ V_2 : &\quad u' = 0, \qquad q' = ku^2, \\ V_3 : &\quad u' = -ku^3, \qquad q' = -ku^2 q. \end{aligned}$$

Thus the boundary equations in antipodal charts differ by a change of sign. All local classifications below concern these two-dimensional restrictions to $w = 0$.

Write

$$S^2_\infty = \{(X, V, F) \in \mathbb{R}^3 : X^2 + V^2 + F^2 = 1\}, \qquad p_f^\pm = (0, 0, \pm 1).$$

Here (X, V, F) represents a direction in the original phase space, and $p_f^\pm$ are the force-axis endpoints.

The charts U_1 and V_1 cover all directions with $X \neq 0$. Neither contains an equilibrium at infinity. In U_2, the line $u = 0$ corresponds to

$$(X, V, F) = \frac{(0, 1, q)}{\sqrt{1 + q^2}}, \qquad q \in \mathbb{R},$$

which gives all directions with $X = 0$ and $V > 0$. The chart V_2 gives the directions with $X = 0$ and $V < 0$. The only remaining points with $X = 0$ are p_f^+ and p_f^-; they are equilibria at the origins of U_3 and V_3. Therefore the complete equilibrium set is the great circle

$$\mathcal{E}_\infty = \{(X, V, F) \in S^2_\infty : X = 0\}. \tag{81}$$

The boundary Jacobians in U_2 and U_3 are, respectively,

$$\begin{pmatrix} 0 & 0 \\ -2ku & 0 \end{pmatrix}, \qquad k \begin{pmatrix} 3u^2 & 0 \\ 2uq & u^2 \end{pmatrix}.$$

Both vanish on $u = 0$, as do their counterparts in the negative charts. Linearization therefore does not determine the local behavior. Instead, we use the explicit solutions of the boundary equations.

4.2. **Local phase portraits.**

Points other than $p_f^{\pm}$. Every such point of $\mathcal{E}_\infty$ has $V \neq 0$, so it can be studied in U_2 or V_2. In U_2, let $(0, q_*)$ be the equilibrium under consideration and put $z = q - q_*$. Then

$$u' = 0, \qquad z' = -ku^2,$$

with solutions

$$u(s) = u_0, \qquad z(s) = z_0 - ku_0^2 s.$$

The line $u = 0$ is stationary. On either side of it, trajectories are straight lines parallel to the equilibrium line, directed toward decreasing z if $k > 0$ and increasing z if $k < 0$. Since u remains a nonzero constant on every moving trajectory, none approaches $(0, q_*)$ in either time direction. The portrait in V_2 is the same with the arrows reversed.
The force-axis endpoints. The point p_f^+ is the origin of U_3. In (80), points with $u = 0$ are stationary. For $u \neq 0$, we have

$$\frac{dq}{du} = \frac{q}{u},$$

so the nonstationary trajectories lie on the rays

$$q = Cu, \qquad C \in \mathbb{R}, \qquad u \neq 0.$$

Indeed, the solution through (u_0, q_0), with $u_0 \neq 0$, is

$$(u(s), q(s)) = \frac{(u_0, q_0)}{\sqrt{1 - 2ku_0^2 s}}, \qquad 1 - 2ku_0^2 s > 0.$$

For $k > 0$, it approaches the origin as $s \to -\infty$ and moves away in forward time. Thus the rays at p_f^+ are outgoing.

At p_f^-, the boundary equations in V_3 have the opposite sign, and their solutions are

$$(u(s), q(s)) = \frac{(u_0, q_0)}{\sqrt{1 + 2ku_0^2 s}}, \qquad 1 + 2ku_0^2 s > 0.$$

For $k > 0$, these rays approach the origin as $s \to +\infty$. They are therefore incoming at p_f^-. When $k < 0$, both orientations reverse. Any divergence of these chart coordinates only marks departure from the chart, not termination of the global compactified orbit.

To determine stability on the sphere, put $r^2 = u^2 + q^2$. Then

$$(r^2)' = \begin{cases} 2ku^2 r^2, & \text{in } U_3, \\ -2ku^2 r^2, & \text{in } V_3. \end{cases}$$

At the endpoint with incoming rays, r cannot increase in forward time. This proves Lyapunov stability. The endpoint is not asymptotically stable, however, because every neighborhood contains other equilibria on $u = 0$, and these points remain fixed rather than converge to the endpoint. At the opposite endpoint, arbitrarily close nonstationary points move away, giving forward instability and Lyapunov stability in reversed time. These are nonisolated equilibria, not isolated nodes or cusps.

4.3. **Global connections on the sphere at infinity.** We now determine how these local trajectories connect on the sphere. Every nonstationary point has $X \neq 0$ and hence lies in U_1 or V_1. First consider U_1. By (76), $u = m$ is constant and $q' = -k$. On the sphere, this trajectory is represented by

$$(X, V, F) = \frac{(1, m, q)}{\sqrt{1 + m^2 + q^2}}, \qquad q \in \mathbb{R}.$$

It lies in the plane $V = mX$. This plane passes through the center of the sphere and the force axis, so its intersection with the sphere is a great circle. The condition $X > 0$ selects one open semicircle. The other semicircle, with $X < 0$, is represented in V_1.

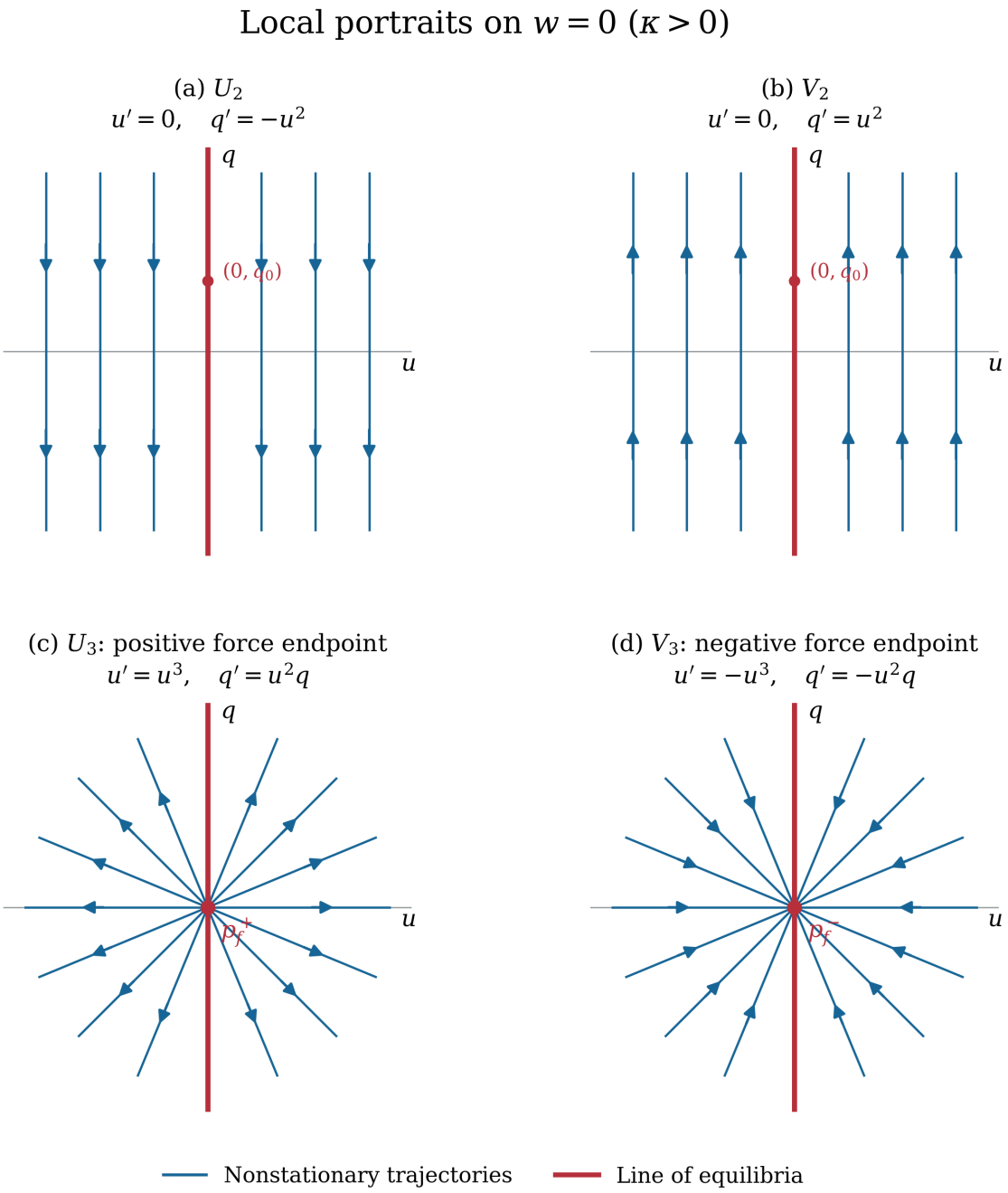


FIGURE 2. Local boundary portraits for $\kappa > 0$, after a positive time rescaling setting $k = 1$. In U_2 and V_2, moving trajectories run parallel to the red equilibrium line. In U_3 and V_3, they form outgoing and incoming rays at p_f^+ and p_f^-, respectively. Every point on a red line is stationary. For $\kappa < 0$, all blue arrows reverse.

Thus all nonstationary orbits can be written as

$$\Gamma_{\delta,m} = \left\{ \frac{\delta(1, m, q)}{\sqrt{1 + m^2 + q^2}} : q \in \mathbb{R} \right\}, \qquad \delta \in \{-1, 1\}, \quad m \in \mathbb{R}.$$

Their endpoints are determined by

$$\lim_{q\to+\infty}\frac{\delta(1,m,q)}{\sqrt{1+m^2+q^2}}=(0,0,\delta),\qquad \lim_{q\to-\infty}\frac{\delta(1,m,q)}{\sqrt{1+m^2+q^2}}=(0,0,-\delta). \tag{82}$$

Hence every nonstationary orbit joins p_f^+ and p_f^-.

If $k>0$, then q decreases in U_1, giving the direction from p_f^+ to p_f^-. In V_1, q increases, but the minus sign in the sphere representation gives the same direction. Therefore

$$\alpha(\Gamma_{\delta,m})=\{p_f^+\},\qquad \omega(\Gamma_{\delta,m})=\{p_f^-\}\qquad (k>0).$$

For $k<0$, the arrows reverse, and

$$\alpha(\Gamma_{\delta,m})=\{p_f^-\},\qquad \omega(\Gamma_{\delta,m})=\{p_f^+\}.$$

These limits are taken in the global compactified flow. The chart time changes preserve orientation, and the smooth flow on the compact sphere is complete. Its nonstationary orbits approach the equilibrium endpoints only in infinite compactified time.

Thus the outgoing rays at one endpoint continue along great semicircles and become incoming rays at the other. Every nonstationary boundary orbit is heteroclinic; no such orbit is periodic or homoclinic. All boundary equations depend only on k, whose magnitude changes the speed and whose sign determines the orientation. Since $\tau>0$, the oriented portrait depends only on the sign of κ.

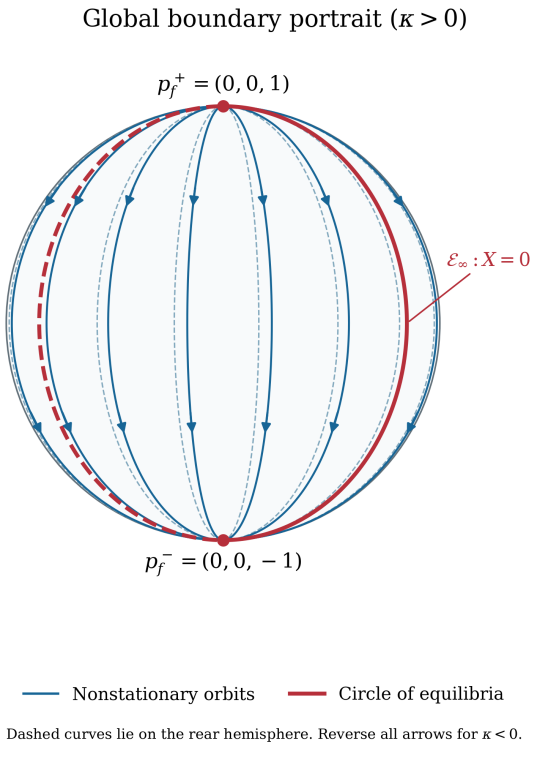


FIGURE 3. Boundary phase portrait for $\kappa>0$. The red great circle $\mathcal{E}_\infty$ consists of equilibria. Every nonstationary orbit is a great semicircle directed from p_f^+ to p_f^-. Dashed curves lie on the rear hemisphere. For $\kappa<0$, all arrows reverse.

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